Chi-Yao Huang

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Education

- **Arizona State University** — Tempe, AZ
  Ph.D. in Computer Science (Advisor: Prof. Yezhou Yang) — *2023 – Present*
  M.S. in Robotics and Autonomous Systems — *2021 – 2023*
- **National Taiwan University** — Taipei, Taiwan
  M.S. in Mechanical Engineering — *2015 – 2017*
- **National Sun Yat-Sen University** — Kaohsiung, Taiwan
  B.S. in Mechanical Engineering — *2010 – 2014*

Publications

- **Domain Expansion: A Latent Space Construction Framework for Multi-Task Learning (ICLR 2026):** Proposed a novel framework to mitigate latent representation collapse in multi-task learning by designing orthogonal pooling mechanism in latent space. Demonstrated superior interpretability and compositional generalization on ShapeNet and MPIIGaze benchmarks. (project | paper | code | talk)
- **VOCAL: Visual Odometry via ContrAstive Learning (WACV 2026):** Reformulated visual state estimation as a label-ranking problem utilizing Bayesian inference and contrastive learning; achieved multimodal compatibility and explainable spatial intelligence on the KITTI benchmark. (project | paper | talk | demo)

Work Experience

- **Arizona State University** — Tempe, AZ
  Ph.D. Student – Prof. Yezhou Yang — *Aug 2021 – Present*
  - **Research:** Developing an interpretable and controllable latent-centric framework that unifies 3D geometry, semantics, and generative modeling, bridging the gap between the real world and AI.
  - **Funding:** Supported by Toyota Research Institute of North America (TRINA).
- **Advanced and Creative Team, HTC VIVE (acquired by Google)** — New Taipei, Taiwan
  Core Member, Team Lambda (VR/AR Innovations) — *Sep 2017 – Feb 2021*
  - **Core Contributions:** Served as a core team member in design reviews, technical planning, and engineering execution for VR/AR products. Mentored junior engineers and contributed to key technical decisions in tracking and spatial computing systems.
  - **VIVE COSMOS:** Architected a multi-camera tracking system achieving 0.4mm precision via robust bundle adjustment; developed a high-performance "mothership" SLAM system for consumer VR. (product | demo)
  - **VIVE FOCUS 3:** Led prototyping for a flagship MR system; optimized a tightly coupled Visual-Inertial SLAM (VIO) pipeline, achieving 4× execution speed-up on resource-constrained embedded systems. (product)
  - **VIVE XR Elite:** Engineered low-latency VIO and obstacle avoidance systems to enhance user safety; designed core data structures for real-time spatial computing in mixed-reality environments. (product)
  - **VIVE FLOW:** Designed an inside-out tracking algorithm for limited-FoV sensors. Developed a prototype motion-tracking pipeline integrating real-time hand tracking. Collaborated with hardware and firmware teams to resolve CPU load and thermal issues in an AR device. (product)
  - **Scene:** Engineered a voxel-based obstacle mapping system for VR safety and integrated semantic maps for real-world interaction.
- **NTU Robotics Lab, National Taiwan University** — Taipei, Taiwan
  Graduate Student and Vice System Manager – Prof. Han-Pang Huang — *Sep 2015 – Jun 2017*
  - **Vice System Manager:** Developed orientation materials and managed robotic equipment including robotic arms and mobile robots.
  - **Research:** Conducted research in SLAM, 3D reconstruction, dense mapping, and robot motion planning.
- **Company of Air Defense Artillery Battalion, R.O.C. Army** — Taiwan
  Battalion Dispatcher — *Sep 2014 – Sep 2015*
  - **Operations:** Managed the dispatch and coordination of military vehicles, trucks, and anti-aircraft systems during training and military exercises.

Projects

- **Aerial Pose Estimation (Sponsored by Toyota Research Institute of North America):** Developed a robust high-altitude pose estimation framework for UAVs using IR-IMU sensor fusion, maintaining reliability under adverse weather and lighting conditions. (demo)
- **3D Landmark Reconstruction for Robot Localization and Mapping:** Implemented a learning-based 3D reconstruction pipeline that jointly optimizes camera trajectories and object-level semantic poses for enhanced robot localization. (demo)
- **3D Reconstruction and Path Planning with Signed Distance Function:** Fused dense image with feature points for efficient SLAM, modified bundle adjustment using Lie groups and quaternions, implemented voxel hashing for dense mapping, and integrated SLAM with biped robot path planning.
- **Humanoid Robot:** Humanoid Robot Control: Developed stable bipedal walking patterns on non-planar surfaces by optimizing ZMP trajectories and implemented a localized A* planner for dynamic obstacle navigation. (demo)
- **Vehicle Speed Control System:** Designed a PID controller for vehicle speed regulation.
- **Stairlift for Elderly People:** Developed a PLC-based stair-climbing system to improve mobility for elderly users.

Honors, Awards & Grants

- **Research Grant, Toyota Research Institute of North America (2023 – Present)**
- **Ira A. Fulton Schools of Engineering Fulton Fellows Award, Arizona State University**
- **Student with Distinction, Arizona State University**

Skills

- **Programming:** Python, C/C++, CUDA, JavaScript, MATLAB
- **Machine Learning:** Deep Learning, Contrastive Learning, Representation Learning, Generative Models, Multi-Task Learning
- **3D Computer Vision:** SLAM, 3D Reconstruction, Bundle Adjustment, Sensor Fusion
- **Robotics & Math:** Spatial Intelligence, Path Planning, Lie Theory, Quaternions, Optimization
- **Tools & Frameworks:** PyTorch, OpenCV, OpenGL, Git, Eigen, Ceres, Docker, ROS, SIMD Optimization, CEVA/Hexagon DSP

Languages

- English (Fluent), Mandarin (Native)